



## Forest Fire Agent-Based Model

# Forest Fire ABM

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Smokers · Firefighters · Spreading Fire · NetLogo · April 2026

# Presentation Outline

ODD Protocol + NetLogo Code + Results

1

## Purpose

Why we built this model and the research question

2

## Entities & State Variables

Patches, Smokers, Firefighters and their properties

3

## Process & Scheduling

What happens each tick and in what order

4

## Design Concepts

Emergence, sensing, interaction, stochasticity

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## Initialization

Setup procedure and the three sliders

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## NetLogo Code

Key procedures with print statement demo

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## Results & Tipping Point

Main findings from the simulation

ODD Step 1

# Purpose & Overview



Why did we build this model?

# Purpose

ODD Step 1

## Research Question

How many firefighters are needed to prevent a forest from burning down, given a fixed number of careless smokers?

## Objectives

Model the competitive dynamics between fire-starters (Smokers) and responders (Firefighters)

Identify the tipping point — the critical ratio where the forest collapses

Show how emergent global behaviour arises from simple local interaction rules

Demonstrate the full ODD protocol applied to an ecological ABM in NetLogo

# Entities & State Variables

ODD Step 2 — Who exists and what properties do they have?



## Patches (Environment)

pcolor: green = healthy forest | red = active fire

Green patches can be ignited by a smoker. Red patches spread fire to one random green neighbour per tick.



## Smokers (Turtles)

xcor, ycor: position

carelessness: ignition probability (= ignition-probability slider)

Walk randomly (rt/lt 60, fd 1). On a green patch: ignite with probability = carelessness.



## Firefighters (Turtles)

xcor, ycor: position

speed = 3 | scan-radius = 4

Scan radius for fire → chase nearest patch → extinguish current + radius-3 area. Patrol randomly if no fire found.



## Globals

initial-trees: total patch count at setup

perc-burned: % of red patches

Updated every tick via update-stats. Used to detect end conditions.

# Process & Scheduling

ODD Step 3 — What happens each tick, in what order?

1

## smoker-walk

All smokers move randomly then probabilistically ignite their current patch

2

## spread-fire-logic

Each red patch spreads fire to one random green neighbour (Moore neighbourhood)

3

## firefighter-go

Each firefighter scans radius, chases nearest fire, extinguishes area

4

## update-stats

perc-burned updated; simulation auto-stops if all red

5

## tick

Clock advances by 1

## Scheduling Notes

Smokers act BEFORE fire spreads — avoids order bias

Fire spreads to exactly ONE neighbour per tick — controlled rate

Firefighters respond AFTER fire spreads — realistic lag

Auto-stop: all burned

# Design Concepts

ODD Step 4 — Emergence, sensing, interaction, stochasticity

## Emergence

Forest survival or collapse is NOT programmed. It emerges from thousands of local ignition, spreading, and extinguishing decisions.

## Sensing

Firefighters sense fire within scan-radius=4. Smokers are completely blind — they walk and drop cigarettes randomly.

## Interaction

Indirect only: smokers and firefighters never meet. Both interact with patches — one turns them red, the other green.

## Adaptation

Firefighters switch between patrol (random walk) and chase (directed movement) depending on whether fire is detected.

## Stochasticity

Ignition is probabilistic (carelessness). Random walk uses  $rt/lt$  random. Fire target neighbour is chosen randomly.

## Observation

Monitor: % burned, tick count. Print statements log every individual ignition and extinguish event in the output console.

# Initialization

ODD Step 5 — What the world looks like at tick 0

```
globals [  
  initial-trees  
  perc-burned  
]  
  
breed [smokers smoker]  
breed [firefighters firefighter]  
  
smokers-own [ carelessness ]  
firefighters-own [ speed scan-radius ]
```

```
to setup  
  setup-environment  
  setup-smokers  
  setup-firefighters  
  print "=== Simulation setup complete ==="  
  print (word "  Trees: " initial-trees)  
  print (word "  Smokers: " num-smokers)  
  print (word "  Firefighters: " num-firefighters)  
  print (word "  Ignition probability: " ignition-probability)  
end
```

```
to setup-environment  
  clear-all  
  ask patches [ set pcolor green ]  
  set initial-trees count patches  
  reset-ticks  
end
```

## Interface Sliders

### num-smokers

Range: 1 – 20 Default: 1

→ interface slider

### num-firefighters

Range: 1 – 20 Default: 1

→ interface slider

### ignition-probability

Range: 0.0–1.0 Default: 0.05

→ interface slider

All patches start green. Agents spawn at random positions. Parameters are fixed per run — set sliders before pressing Setup.

# NetLogo Code — Smoker Logic

How smokers move and start fires

```
to setup-smokers
  create-smokers num-smokers [
    setxy random-xcor random-ycor
    set shape "person"
    set color black
    set size 1.5
    set carelessness ignition-probability
    print (word "Smoker created at (" (precision xcor 1) ", " (precision ycor 1) ") carelessness=" (precision carelessness 2))
  ]
end

to smoker-walk
  ask smokers [
    rt random 60
    lt random 60
    fd 1
    if pcolor = green [
      if random-float 1 < carelessness [
        set pcolor red
        print (word "Smoker at (" (precision xcor 1) ", " (precision ycor 1) ") dropped cigarette! Patch ignited.")
      ]
    ]
  ]
end
```

# NetLogo Code — Firefighter Logic

How firefighters detect, chase, and extinguish fire

```
to setup-firefighters
  create-firefighters num-firefighters [
    setxy random-xcor random-ycor
    set shape "car"
    set color blue
    set size 1.8
    set speed 3
    set scan-radius 4
    print (word "Firefighter created at (" (precision xcor 1) ", " (precision
ycor 1) ") scan-radius=" scan-radius)
  ]
end
```

```
to firefighter-go
  ask firefighters [
    let fire-patches patches in-radius scan-radius with [pcolor = red]
    ifelse any? fire-patches [
      let target min-one-of fire-patches [distance myself]
      face target
      fd min (list speed (distance target))
    ]
  ]
end
```

```
    ; Extinguish fire on current patch
    if pcolor = red [
      set pcolor green
      print (word "Firefighter at (" (precision xcor 1) ", " (precision
ycor 1) ") extinguished fire!")
    ]
  ]
end
```

```
    ; Also extinguish all red patches within reach
    ask patches in-radius 3 with [pcolor = red] [
      set pcolor green
      print (word " -> Nearby patch at (" pxcor ", " pycor ") also
extinguished")
    ]
  ]
end
```

```
    ; No fire in scan radius - move toward any fire
    on the whole map
    let any-fire patches with [pcolor = red]
    ifelse any? any-fire [
      face min-one-of any-fire [distance myself]
      fd speed
    ] [
      rt random 45
      lt random 45
      fd 1
    ]
  ]
end
```

# Demonstration — Print Statements

Tracing individual agent interactions in the NetLogo output console

```
to setup
  setup-environment
  setup-smokers
  setup-firefighters
  print "=== Simulation setup complete ==="
  print (word " Trees: " initial-trees)
  print (word " Smokers: " num-smokers)
  print (word " Firefighters: " num-firefighters)
  print (word " Ignition probability: " ignition-probability)
end

to go
  smoker-walk
  spread-fire-logic
  firefighter-go
  update-stats

  if not any? patches with [pcolor = green] [
    print "=== FOREST FULLY BURNED - simulation stopped ==="
    stop
  ]

  tick
end
```

## What to observe

### Orange — Smoker events

*Smoker created at (x, y) carelessness=0.10"*  
*"Smoker at (x, y) dropped cigarette! Patch ignited.*

### Green — Firefighter events

*Firefighter created at (x, y) scan-radius=4"*  
*"Firefighter at (x, y) extinguished fire!"*  
*"-> Nearby patch at (x, y) also extinguished*

### Teal — System events

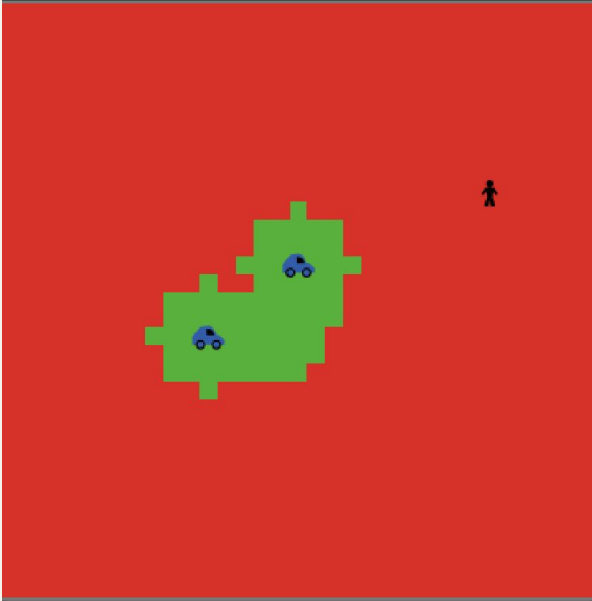
*=== Simulation setup complete ==="*  
*"=== FOREST FULLY BURNED - simulation stopped*  
*===*

### Red — Active fire

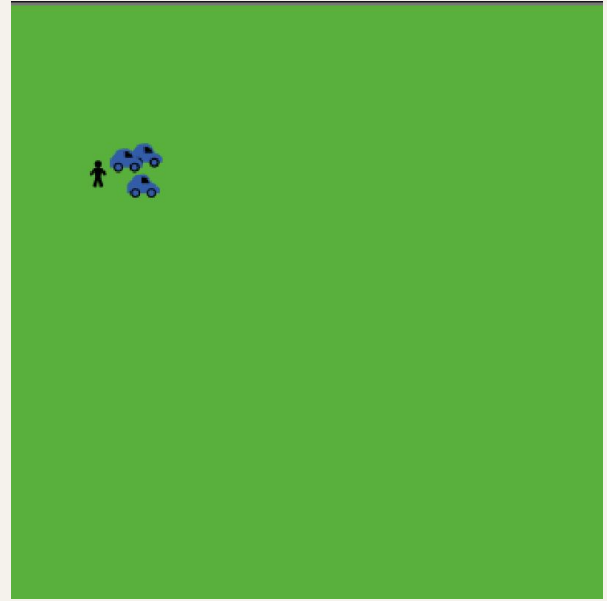
Visible in the world view as red patches spreading across the grid

# Results

How many firefighters does the forest need to survive?



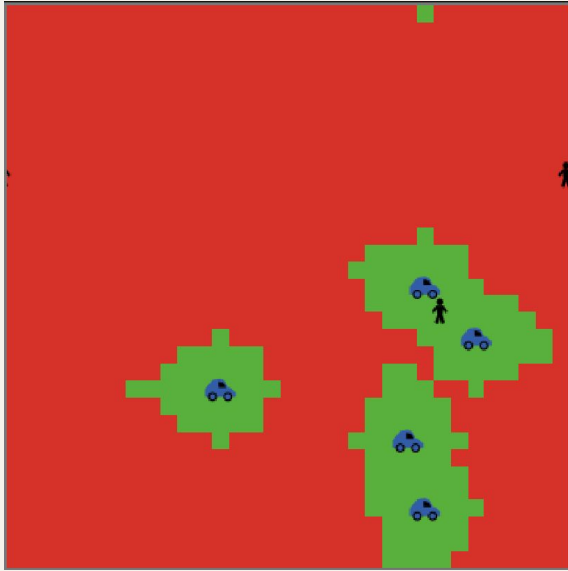
Smoker: 1  
Firefighter:2  
ignition-probability: 0.05  
Ticks:1440  
Perc Burned: 93.9



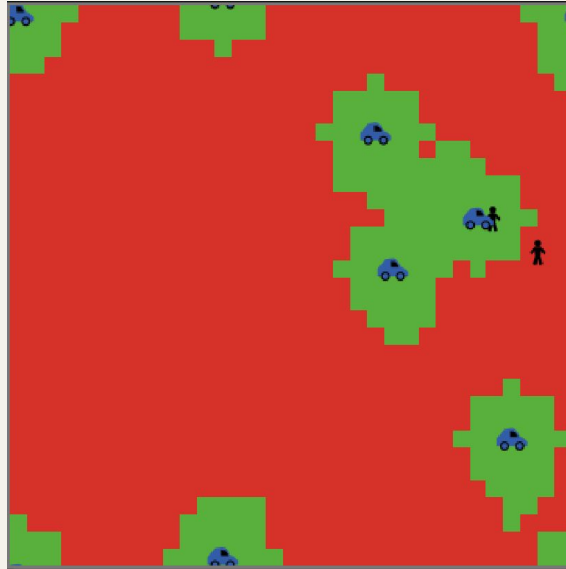
Smoker: 1  
Firefighter: 3  
ignition-probability: 0.05  
Ticks: 10233  
Perc Burned: 0

# Results — Increasing firefighters

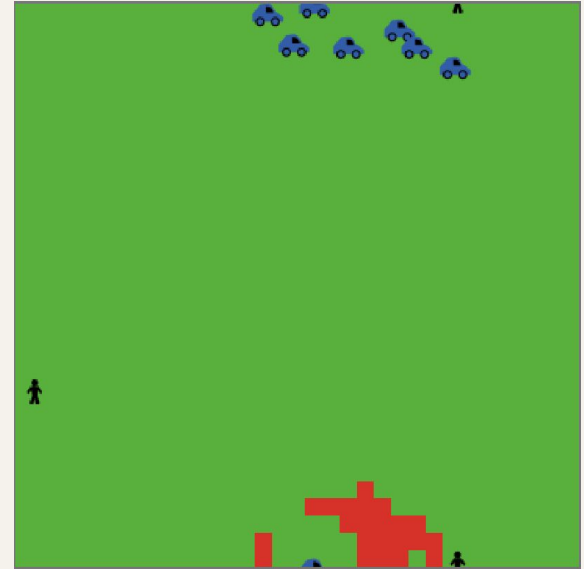
How many firefighters does the forest need to survive?



Smoker: 2  
Firefighter: 5  
ignition-probability: 0.05  
Ticks:2148  
Perc Burned: 85.9



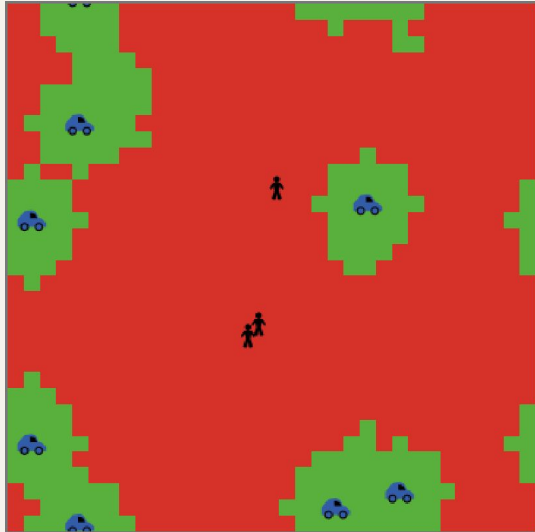
Smoker: 2  
Firefighter: 6  
ignition-probability: 0.05  
Ticks: 7254  
Perc Burned: 81



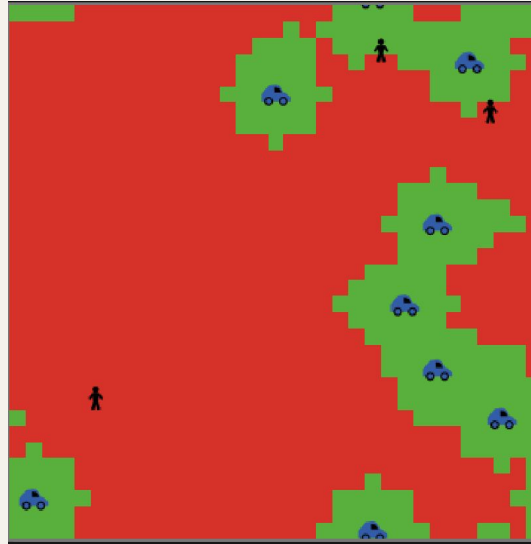
Smoker: 2  
Firefighter: 7  
ignition-probability: 0.05  
Ticks: 10220  
Perc Burned: 2

# Results —

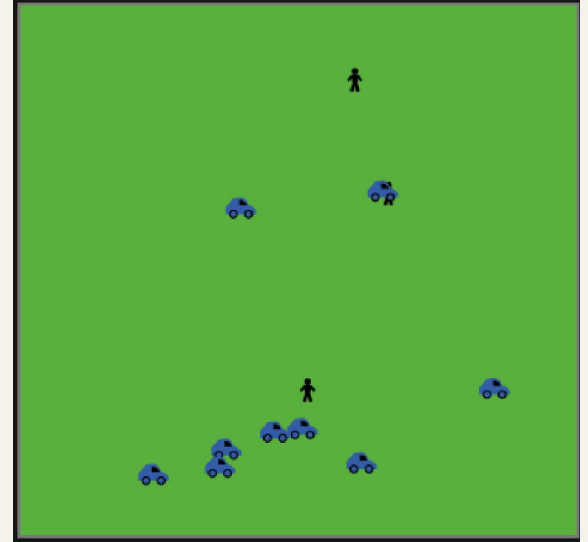
How many firefighters does the forest need to survive?



Smoker: 3  
Firefighter: 7  
ignition-probability: 0.05  
Ticks: 4971  
Perc Burned: 78.3



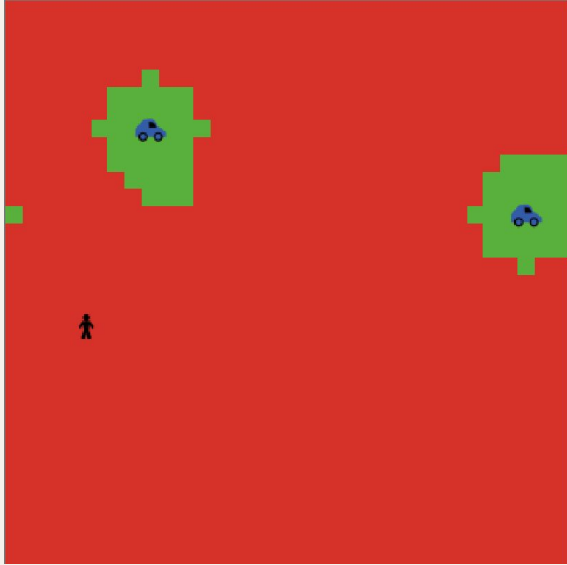
Smoker: 3  
Firefighter: 8  
ignition-probability: 0.05  
Ticks: 5243  
Perc Burned: 76



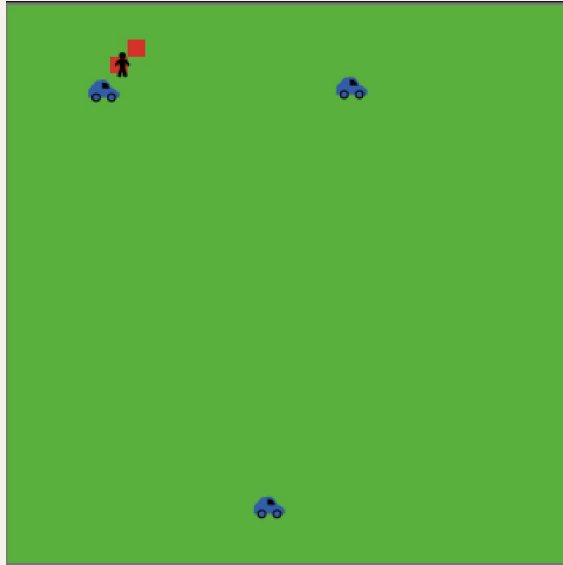
Smoker: 3  
Firefighter: 9  
ignition-probability: 0.05  
Ticks: 10170  
Perc Burned: 0

# Results

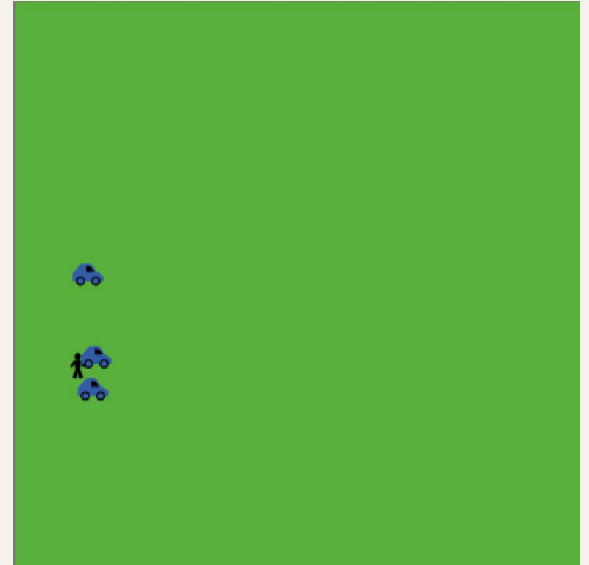
How many firefighters does the forest need to survive?



Smoker: 1  
Firefighter: 2  
ignition-probability: 0.10  
Ticks: 581  
Perc Burned: 93.8



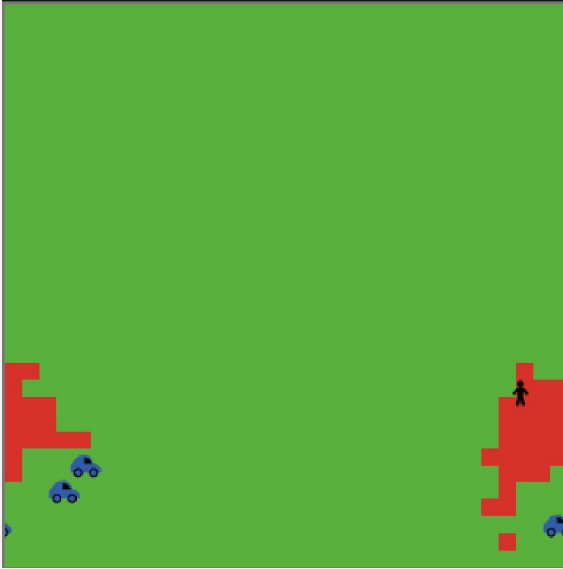
Smoker: 1  
Firefighter: 3  
ignition-probability: 0.10  
Ticks: 10190  
Perc Burned: 0.2



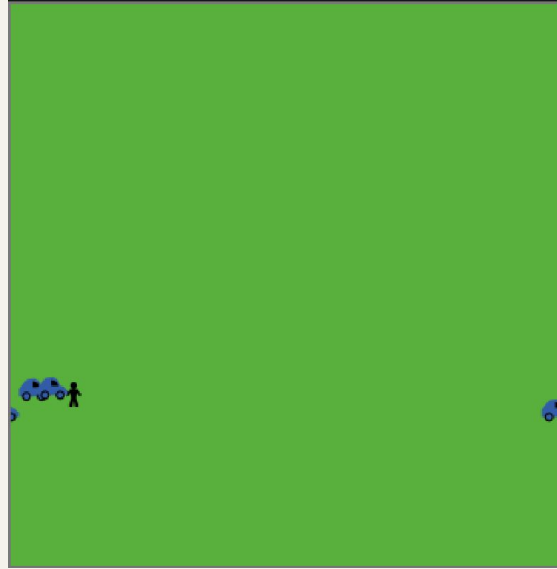
Smoker: 1  
Firefighter: 3  
ignition-probability: 0.15  
Ticks: 10366  
Perc Burned: 0

# Results

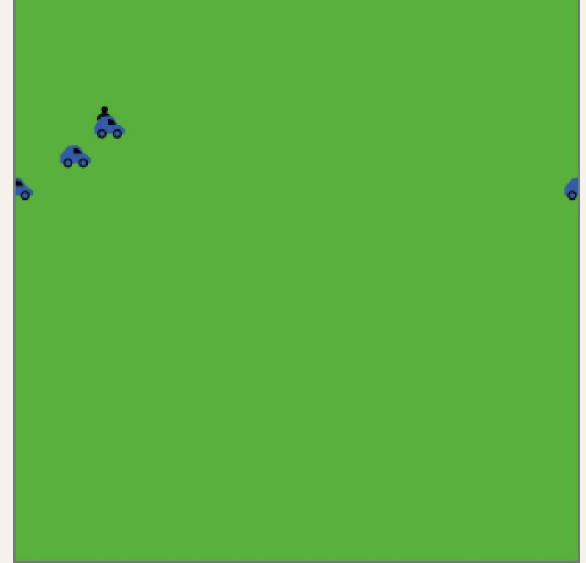
How many firefighters does the forest need to survive?



Smoker: 1  
Firefighter: 2  
ignition-probability: 0.20  
Ticks: 581  
Perc Burned: 93.8



Smoker: 1  
Firefighter: 3  
ignition-probability: 0.30  
Ticks: 10080  
Perc Burned: 0



Smoker: 1  
Firefighter: 3  
ignition-probability: 0.50  
Ticks: 10662  
Perc Burned: 0

# Results — Key Findings

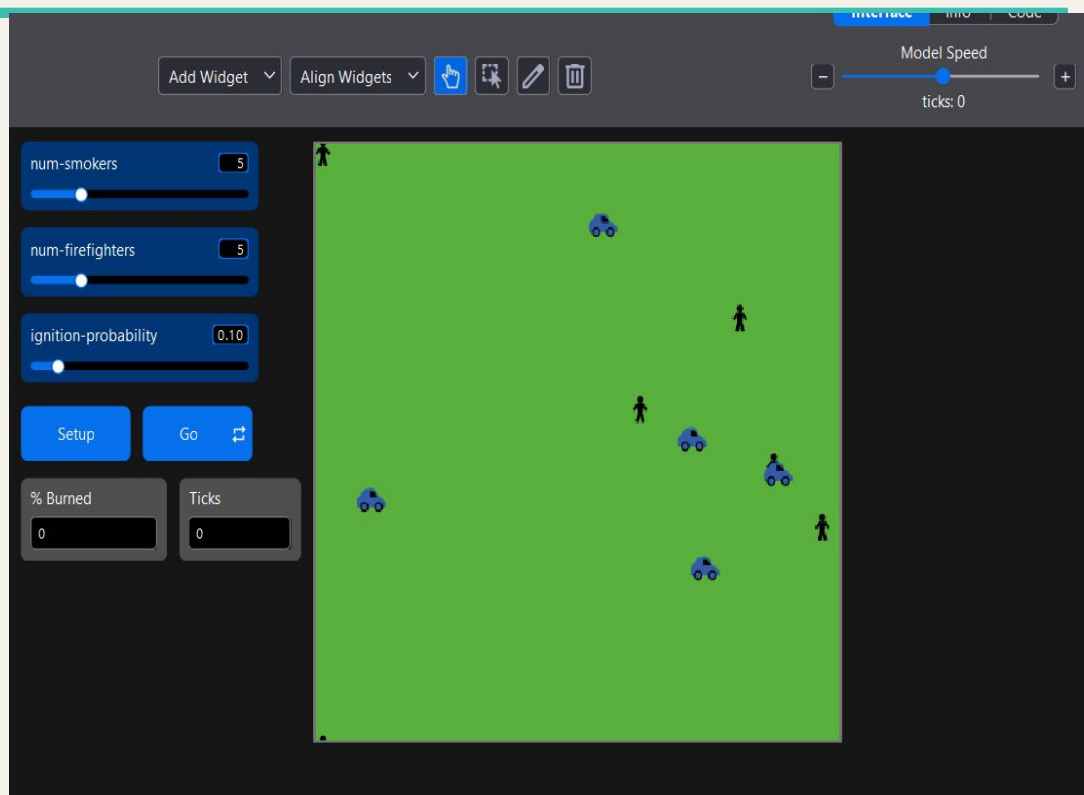
How many firefighters does the forest need to survive?

**Tipping point  $\approx 3$  firefighters per smoker, consistent across 1, 2 and 3 smokers**

**Below tipping point: forest burns around 5000 ticks**

**Above tipping point: forest survives with 0–2% burned after 10000+ ticks**

**Higher ignition-probability (up to 0.50) does NOT change the ratio needed (if there is a redundancy of firefighters)**



Q&A TIME